Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/09/2010
PI Name:	Straume, Tore Ph.D.		
Project Title:	Lunar EVA Dosimetry: Small Active Dosimetry System for Lunar Extravehicular Activity Missions: Spacesuit and Tool-Box Applications		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:	**TechPort:**		Yes
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **ARS**:Risk of Acute Radiation Syndromes Due to Solar Particle Events (SPEs)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-3943
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Division		
PI Address 2:	MS236-7		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	11/01/2007	**End Date:**	10/31/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: title changed per NSBRI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas Engineering Experiment Station) Lusby, Terry (NASA ARC)		
Grant/Contract No.:	NCC 9-58-RE01302		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The aim of the project is the development of a compact active dosimetry system that can provide real-time radiation monitoring beyond low-Earth orbit (LEO). The dosimetry system would be capable of active, real-time monitoring during both ambient and solar particle event (SPE) conditions. It will be designed to be responsive to the doses, dose rates, and radiation qualities (including secondary neutrons) expected in interplanetary space. As astronauts travel beyond LEO and explore asteroids, the Moon, and Mars, there will be a critical need for active personal radiation monitoring. Of particular concern is the need for real-time radiation dosimetry information during SPEs so that astronauts can seek prompt shelter. SPEs occur unexpectedly and with highly variable dose rate and can result in a broad range of doses to astronauts. Also, the quality of the radiation, both primary and secondary, can vary significantly between and within SPEs. It is expected that secondary radiations produced in the spacecraft or in the surface regolith by incident SPE particles, mostly protons, may result in significant neutron contributions to the dose equivalent. This dynamic and complex radiation environment requires the development of suitable detection systems. This project is a joint effort between NASA Ames Research Center (ARC), Colorado State University (CSU), and Texas A&M University (TAMU). CSU designs and fabricates the TEPC sensors, TAMU designs and builds the pre-amplifier with charge integrator, and ARC designs and builds the miniaturized electronics package with bias voltage power supply and microprocessor and programs the microprocessor to calculate variance-covariance, dose and dose-equivalent. Progress during the third year: The Mod 1 versions of these components were integrated with the tissue-equivalent sensors fabricated by CSU and tested using ground based simulated space radiation. The Mod 1 monitor was exposed to 226 MeV protons, 200 MeV/n C, and 382 MeV/n Fe. The focus of this initial Mod-1 test was to (1) determine if the detector response behaved as expected for low and high LET radiations, (2) determine if the level of electronic noise in the combined system was low enough to permit the detection of high-energy protons, and (3) and look at the angular dependence of the single wire detector. Measurements were made at four beam angles and frequency mean lineal energy (YF) and dose mean lineal energy (YD) calculated. The results demonstrated that minimally ionizing protons (0.22 keV/μm) are detectable. The results do not show detectable angular dependence within the uncertainties of the measurements. A higher than expected response was observed in the mid LET range for Fe, which is being addressed in Mod 2, and includes improved sensor design, improved low noise pre-amp design with built-in integrator circuit, and improved miniaturized electronics package with data processing. Mod 2 will be tested during coming months. Mod-2 also include variance-covariance capability to obtain radiation quality using a single TEPC detector. Various single and multiwire sensors, pre-amp mods, and electronics components are being tested and compared for their performance. The optimum sensor, pre-amp, and electronics package will be selected for the final prototype. Initial tests using a neutron source were made of the variance-covariance method and results show that the monitor operated as planned. Proposed research plan for the coming year: Complete development and fine tuning of the pre-amplifier and microelectronics package. This includes completing and testing variance-covariance, completing and testing final miniaturized electronics package with low noise, well-regulated dc power supply, complete modifications and testing of pre-amp charge integrator, and make selection of final components for prototype monitor. This would be followed by performance testing the final prototype assembly, which would include the jointly (ARC, CSU, TAMU) selected Mod 2 sensor designs. The source tests will include NSRL and Chiba facilities, as well as proton tests at the 88" cyclotron at Berkeley.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The compact real-time radiation monitoring capability developed in this project may be useful in homeland security and DOD applications, where first responders would require small personal dosimeters capable of detecting real-time mixed gamma and neutron radiations.		

Task Progress:	During year 3 we developed and tested our Mod-1 prototype using simulated space radiation. The detector was exposed to 226 MeV protons, 200 MeV/n C, and 382 MeV/n Fe. The focus of this initial Mod-1 test was to (1) determine if the detector response behaved as expected for low and high LET radiations, (2) determine if the level of electronic noise in the combined system was low enough to permit the detection of high-energy protons, and (3) and look at the angular dependence of the single wire detector. Measurements were made at four beam angles and frequency mean lineal energy (YF) and dose mean lineal energy (YD) calculated. The results for protons were calculated for lineal energies above 0.22 keV/µm and demonstrated that they are detectable with Mod 1. The results do not show detectable angular dependence within the uncertainties of the measurements. We observed a higher than expected response in the mid LET range for Fe. This is being addressed in Mod 2, which will be tested during the next few months. We have also performed initial tests of variance-covariance; results indicate that the charge integrator is performing as expected. During year 3 we also developed Mod-2 prototypes of the pre-amplifier and electronics package. This includes improved low noise pre-amp design with built-in integrator circuit, and miniaturized electronics package with data processing. Mod-2 includes variance-covariance capability to obtain radiation quality using a single TEPC detector. Various single and multiwire sensors, pre-amp mods, and electronics components were also developed and will be tested and compared for their performance during year 4. The optimum combination of sensor, pre amp, and electronics package will be selected for the final prototype to be completed by the end of year 4.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Abstracts for Journals and Proceedings	Borak T, Braby L, Heilbronn L, Iwata Y, Lusby T, Murakami T, Oertli D, Straume T, Warner D. "Design and testing of compact tissue equivalent proportional counters for astronauts during extra vehicular activity and inside spacecraft." 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. 56th Annual Meeting of the Radiation Research Society, Abstract Book, September 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=359561f3-493c-4d5c-b014-47e677719c70&cKey=ce8badf9-5f7c-4b65-acf2-01039e0dff25&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Abstracts for Journals and Proceedings	Straume T, Borak T, Braby L, Lusby T, Warner D, Tran H, Oertli D, Liu H, Murakami T, Iwata Y. "Active EVA radiation monitor." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. 21st Annual NASA Space Radiation Investigators' Workshop, Abstract Book, May 2010. , May-2010